Lab Session 15: Final Exam

Objective

To apply numerical methods for interpolation, curve fitting, and numerical integration, using Python in Google Colab only for simple arithmetic support.

Instructions

- This exam has two parts: Interpolation & Curve Fitting and Numerical Integration.
- Show essential steps for each calculation.
- You may use Google Colab; keep code minimal and only for basic arithmetic.
- Do NOT use automated methods, regression libraries, or integration tools.
- Keep explanations concise (2–3 sentences where required).

Part A - Interpolation and Curve Fitting

You are given the concentration data (mg/L) at different times (hours):

t (h)	0	1	2	3
C (mg/L)	10.0	7.8	5.9	5.1

1. **Lagrange Interpolation**
   Briefly describe how a Lagrange interpolation polynomial is formed. Then estimate **P (2.5)** using simple manual steps or minimal Python arithmetic (no interpolation libraries).

2. **Linear Curve Fitting**
   Using basic formulas for linear least-squares fitting:
   - Compute the slope ***m*** and intercept ***c***.
   - Estimate **C (2.5)** using the fitted line.

3. **Comparison**
   In 2–3 sentences, explain which estimate (interpolation or linear fit) is more reliable for this dataset. Mention **one limitation of interpolation** and **one advantage of linear fitting**.

Part B - Numerical Integration (Trapezoidal vs Simpson's 1/3 Rule)

A drone's velocity is recorded every second:

t (s)	0	1	2	3	4
v(t) (m/s)	0	3.0	5.5	6.8	7.7

1. **Trapezoidal Rule**

   Use the Trapezoidal Rule to approximate D = $\int_0^3 v\,(t)$ dt. Write the formula and give the final value.

2. **Simpson's 1/3 Rule**

Use Simpson's 1/3 Rule to approximate the same integral. Write the weighted-sum formula and the final answer.

3. **Comparison**

In 2-3 sentences, explain which method is more accurate for this smooth dataset, and in what situations the trapezoidal rule may still be preferred.

Deliverables

Students must submit **one Word document** containing:

1. **A short handwritten or typed solution** with formulas and brief steps.
2. **Python code** (basic arithmetic only) pasted as text.
3. **No advanced numerical libraries** and **no plots**.

Submit the Word file through the **Google Form**.

Learning Outcomes

By completing this exam, students will be able to:

• Construct interpolation and curve-fitting estimates from data.
• Apply numerical integration methods to discrete datasets.
• Compare numerical methods with respect to accuracy and suitability.

Criteria	Achieved (1)	Not Achieved (0)	Marks
Method Understanding	Shows basic understanding of the required numerical methods.	Explanation missing or incorrect.	
Correct Procedure	Applies appropriate formulas or steps correctly.	Uses wrong method or incorrect steps.	
Calculation Accuracy	Numerical results are correct or nearly correct.	Major errors in calculation.	
Clarity of Work	Steps are clear, organized, and easy to follow.	Work is unclear, incomplete, or disorganized.	
Conceptual Explanation	Provides brief, relevant comparison or reasoning.	Explanation missing or irrelevant.	
Weighted CLO Score			
Remarks and Signature			